

Ayush Verma

Software Development Engineer

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Education

SRM University, Andhra Pradesh

B.Tech in Computer Science and Engineering

2023 – 2027

CGPA: 8.6/10.0

Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Work Experience

Purple Bean Agro

June 2025 – July 2025

Full Stack Developer Intern

- Developed 12+ RESTful API endpoints for auth, order, and checkout workflows in Node.js/Express, optimizing query latency by restructuring compound MongoDB indexes and aggregation pipelines.
- Built modular JWT authentication with 3-tier Role-Based Access Control (RBAC) and reusable validation middleware, eliminating duplicate parameter checks across mutating checkout endpoints.
- Integrated Razorpay payment gateway with cryptographic HMAC SHA-256 signature verification and idempotent confirmation handling, preventing duplicate charges under network retry conditions.

Projects

LedgerTrace – Event-Sourced Double-Entry Ledger | *Node.js, Express, PostgreSQL, React*

[\[Project Link\]](#)

- Engineered an event-sourced ledger in Node.js & PostgreSQL, deriving balances from an append-only event stream instead of mutable state to ensure auditability and point-in-time state replay.
- Implemented atomic double-entry transfers and compensating reversal chains; benchmarked under 500 concurrent connections at 269 req/sec with zero double-spends and \$0 balance drift.
- Built an independent reconciliation engine validating global debits/credits from sequence zero, alongside an authenticated Server-Sent Events (SSE) live feed; covered with 20/20 test suites.

HireHeaven – AI Job Portal | *Next.js 16, TypeScript, Node.js, Kafka, Redis, PostgreSQL*

[\[Project Link\]](#)

- Architected a 5-service backend in an npm workspaces monorepo behind an API Gateway, implementing JWT access/refresh rotation, Redis token revocation, and strict Zod validation on mutating endpoints.
- Built asynchronous event pipelines with Apache Kafka and Redis for cache-aside reads, explicit write invalidation, and sliding-window rate limiting on critical auth/payment endpoints.
- Shipped end-to-end observability with structured Pino logging and Prometheus `/metrics`; automated full-stack build, typechecking, and Vitest test suites on pull requests via GitHub Actions CI/CD.

Attendy – Real-Time Biometric Attendance | *FastAPI, Python, PostgreSQL, pgvector, WebSockets*

[\[Project Link\]](#)

- Built an asynchronous face recognition service using FastAPI and PostgreSQL `pgvector` with HNSW cosine similarity on 512-d ArcFace embeddings, reducing p50 search latency to 432ms (3.9x speedup).
- Engineered bidirectional WebSocket streaming with Hungarian algorithm identity tracking, 4-of-6 frame temporal consensus, and bounding-box motion heuristics to enforce anti-spoof liveness verification.
- Guaranteed exactly-once concurrency constraints via PostgreSQL partial unique indexes, stress-tested with 50 concurrent threads at 264 req/sec; automated testing with pytest and Vitest in GitHub Actions.

Open Source Contributions

- **Novu (JS/React/SSR):** Fixed SSR compatibility in the SDK by preventing DOM-dependent NovuUI mounting during server rendering and dynamically loading the UI module on the client. [\[PR #12311\]](#)
- **Directus (TypeScript/Data Integrity):** Resolved duplicate relational updates in content-version translation workflows by isolating keyed updates while preserving keyless upserts; added regression coverage. [\[PR #28196\]](#)
- **LibreChat (AI/Backend):** Diagnosed and fixed a fine-tuned endpoint parameter-forwarding bug affecting Flux image generation pipelines, merging upstream into main. [\[PR #15336\]](#)

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, C++, SQL

Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3

Backend & Cloud: Node.js, Express.js, FastAPI, RESTful APIs, AWS, Microservices, WebSockets, Kafka, Docker

Databases & Caching: PostgreSQL, MongoDB, Redis (Cache-Aside, Rate Limiting), pgvector (Vector Search)

System Design: Event Sourcing, Distributed Systems, Concurrency Control, Idempotency, Load Testing, Zero-Drift Replay

DevOps & Testing: Git, GitHub Actions, CI/CD, Vitest, pytest, Prometheus, Grafana, Pino Structured Logging, Postman

Achievements & Honors

- Solved **350+ Data Structures & Algorithms problems** on [LeetCode](#), demonstrating strong problem-solving.
- Finalist at **SDG Hackathon**, **HackSRM 7.0**, and **MSC 2.0 Hackathon**.
- Volunteer, **GDG On Campus SRM AP**: Organized technical workshops and hackathons for 500+ students.

Certifications

- **SAP Certified Generative AI Developer – SAP**

[\[Credential\]](#)

- **MongoDB Associate Developer – MongoDB**

[\[Credential\]](#)